

Dhaval Patel

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EDUCATION

University of Massachusetts Amherst – Amherst, MA

September 2025 – May 2027

Master of Science in Computer Science

GPA: 3.76/4.0

Relevant Coursework - Advance Natural Language Processing, Machine Learning, Neural Networks, Software Engineering

Charotar University of Science and Technology – Anand, India

October 2021 – May 2025

Bachelor of Technology in Computer Engineering

GPA: 9.0/10

Relevant Coursework - Data Structures, Database Management System, Operating System, AI, Image Processing

TECHNICAL SKILLS

Programming Languages: Python, R, C++, HTML, CSS, SQL, Javascript, Node.js

Data Science and Machine Learning: Deep Learning, NLP, Large Language Models (LLMs), RAG (Retrieval Augmented Generation), Data Analysis, Data Visualization, Statistics, Predictive Modeling, Time Series Analysis, Data Wrangling, Feature Engineering, Model Deployment, Hyperparameter Tuning, Cross-Validation, Prompt Engineering

Tools & Technologies: Git, Docker, Amazon Web Services (EC2, S3), Microsoft Azure, Claude Code

Libraries and Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras, Transformers, FastAPI, LangChain, OpenAI, OpenCV, NLTK

PROFESSIONAL EXPERIENCE

Cilans System

October 2024 - June 2025

AI/ML Engineer

Ahmedabad, India

- Architected and deployed scalable backend pipelines for the **GranthAI** document processing platform, integrating multiple third-party APIs (OpenAI, Gemini) to handle high-volume document ingestion and processing.
- Designed and integrated Azure Cosmos DB with scalable indexing and retrieval workflows, supporting multilingual document search at scale with sub-second query response times.
- Reduced end-to-end system latency by 47% by optimizing API call orchestration, implementing efficient caching strategies, and refactoring monolithic workflows into modular components.
- Built RESTful APIs and microservices using FastAPI for document parsing, summarization, and QA functionality, ensuring high availability, modular architecture, and seamless downstream integration with frontend services.
- Collaborated in an Agile environment with cross-functional teams, owning feature delivery from design to deployment using Git workflows, peer code reviews, Docker-based containerization, and CI/CD pipelines for reliable, reproducible releases.

Charotar University of Science and Technology (CHARUSAT)

May 2024 - July 2024

Research Intern

Anand, India

- Streamlined construction resource estimation accuracy by 24% by extracting dimensional information from floor plan images using OCR and optimizing material requirement calculations for greater precision and efficiency.
- Co-authored and published 3 research papers on applied deep learning, presenting findings on BERT-based cognitive skill evaluation, YOLOv8-driven real-time activity detection, and novel approaches to improving model accuracy, efficiency, and real-world applicability across educational and sports analytics domains.

PROJECTS

Sales Management Platform — [Link]

- Distributors used to manually record every sale, stock entry, and membership detail in notebooks — a time-consuming and error-prone process. This platform digitalizes that entire workflow, making it faster, accurate, and accessible in one place.
- Developed end-to-end with Next.js 14, Supabase, TypeScript, and Tailwind CSS with modules for sales, inventory, center sales, membership tracking, and dashboard analytics including low-stock alerts and volume point tracking.
- Generates period reports for profits, revenue, and inventory with printable summaries to help distributors make informed business decisions — developed with the assistance of Claude Code, Anthropic's AI-powered coding tool.

Robust Anomaly Detection under Real-World Image Corruptions — [Link]

- Investigated robustness of industrial anomaly detection models (PaDiM, PatchCore) built on Wide-ResNet-50 against real-world image corruptions such as Gaussian noise, blur, brightness, JPEG across 5 severity levels on the MVTec AD dataset, revealing critical performance gaps where image-level ROC-AUC collapsed to 0.36 below random under brightness corruption for texture categories.
- Designed a corruption-aware data augmentation pipeline that improved model robustness by up to 30% (lifting Gaussian noise AUC from 0.61 to 0.79 and JPEG AUC from 0.55 to 0.78) by training memory banks and Gaussian distributions on mildly corrupted normal samples in PyTorch.

PUBLICATIONS

- **Impact of BERT on Evaluating the Quality of Question Paper Using Bloom's Taxonomy - ICT4SD-2024**
DOI: 10.1007/978-981-97-8526-1_6
- **Real-Time Sports Analysis: Integrating YOLOv8 with Enhanced Feature Pyramid Networks for Volleyball and Cricket Activity Detection - ADCIS-2024** DOI: 10.1007/978-981-96-4536-7_9